

THE UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY



Kiểm thử phần mềm - 22KTPM3
PERFORMANCE TESTING - HW6

Thành Viên

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Chapter 1

Group Information

Group ID: 07

Member Name	Student ID	Assigned Features	Status
Cao Uyển Nhi	22127310	- Homepage- Category- Product	Done
Lưu Thanh Thuý	22127410	- Register	Done
Nguyễn Phước Minh Trí	22127424	- Homepage- Sort- Search	Done
Võ Lê Việt Tú	22127435	- Sign In- Invoice	Done
Trần Thị Cát Tường	22127444	- Homepage- Contact	Done

Chapter 2

Server / PC Configuration

Component	Specification
OS	Windows 11 Pro 64-bit
CPU	Intel Core i7-11800H, 8 cores
RAM	16 GB
Disk	512 GB SSD
JMeter Version	Apache JMeter 5.6.3
Java Version	OpenJDK 17
Browser	Chrome 124 (recording)
Website Under Test	The Toolshop (Localhost, Angular Frontend + REST API Backend)

Chapter 3

Performance Testing

Three performance testing techniques were applied:

- **Load Testing** – Verify the system can handle expected normal traffic (50 users).
- **Spike Testing** – Measure system behavior during sudden load surges.
- **Stress Testing** – Identify the system's maximum capacity and behavior beyond limits.

3.1 Test Case 1: Load Testing

Objective: Verify the system can process 50 concurrent users without performance degradation.

User Journey:

1. Access Homepage (/)
2. Sort products by data from CSV
3. Search for products using a keyword (data from CSV)

Thread Group Configuration:

Parameter	Value
Number of Threads	50
Ramp-up Period	10s
Loop Count	1
Duration	300s
Assertions	Response time 2000 ms, HTTP status 200

Expected Result: - Average response time 2 seconds - Zero or negligible error rate

3.2 Test Case 2: Spike Testing

Objective: Evaluate the system's reaction and recovery time during sudden load increases.

User Journey: Same as Load Testing.

Ultimate Thread Group Configuration:

Stage	Virtual Users	Ramp-up	Hold Load	Shutdown
Warm-up	50	5 sec	30 sec	5 sec
Spike 1	200	5 sec	30 sec	5 sec
Spike 2	400	5 sec	30 sec	5 sec
Spike 3	800	5 sec	30 sec	5 sec
Peak	1000	5 sec	30 sec	5 sec
Recovery	50	5 sec	30 sec	5 sec

Expected Result: - Response time increases but stays < 3000 ms during peaks

- Near-zero error rate
- System returns to normal after load decreases

3.3 Test Case 3: Stress Testing

Objective: Determine the maximum load the system can handle.

User Journey: Same as Load Testing.

Ultimate Thread Group Configuration:

Stage	Virtual Users	Ramp-up	Hold Load	Shutdown
1	10	10 sec	60 sec	5 sec
2	50	20 sec	60 sec	5 sec
3	100	30 sec	60 sec	5 sec
4	200	40 sec	60 sec	5 sec

Expected Result: - Performance starts to degrade when exceeding ~180 users

- At 200 users, noticeable latency and possible failures appear

Chapter 4

Applying AI in Performance Testing

4.1 Prompt 1 – CSV & Data-driven Testing

- **Objective:** Fix CSV parsing issues when values contain commas.
- **Prompt Example:** “CSV contains comma-separated values in a single field; how to configure JMeter to read them correctly?”
- **AI Output:** Suggested changing the delimiter to ; and configuring CSV Data Set Config accordingly.

4.2 Prompt 2 – Ultimate Thread Group Config

- **Objective:** Optimize parameters for Load, Stress, and Spike tests.
- **Prompt Example:** “How to configure Ultimate Thread Group for 50-user load test, 1000-user spike, and 200-user stress test?”
- **AI Output:** Provided detailed ramp-up, hold, and shutdown configurations.

4.3 Prompt 3 – Debugging URI Error

- **Objective:** Fix URISyntaxException caused by special characters in dynamic URLs.
- **Prompt Example:** “How to handle URISyntaxException in JMeter when using CSV variables in URLs?”
- **AI Output:** Suggested removing quotes or encoding commas as %2C.

Chapter 5

Youtube Video Link

- Demo video of Load Testing: [Video demo Load Testing](#)
- Demo video of Spike Testing: [Video demo Spike Testing](#)
- Demo video of Stress Testing: [Video demo Stress Testing](#)

Chapter 6

Self-Evaluation

Criteria	Max Point	Self-assessment
1. Load Testing	3.0	3.0
1.1. Report	1.0	1.0
1.2 Test cases + Bug Report	0.5	0.5
1.3. Scripts, Data	0.5	0.5
1.4 Video	1.0	1.0
2. Stress Testing	3.0	3.0
2.1. Report	1.0	1.0
2.2 Test cases + Bug Report	0.5	0.5
2.3. Scripts, Data	0.5	0.5
2.4 Video	1.0	1.0
3. Spike Testing	3.0	3.0
3.1. Report	1.0	1.0
3.2 Test cases + Bug Report	0.5	0.5
3.3. Scripts, Data	0.5	0.5
3.4 Video	1.0	1.0
Use of AI Tools	1.0	1.0
Total	10.0	10.0